CV Creation using LLMs (Capstone Project Report)

1. Introduction

This project presents an AI-based system that automates resume creation, tailoring, and optimization using Large Language Models (LLMs). It focuses on improving resume quality and aligning it with job requirements for better ATS compatibility.

2. Objective

1. Extract structured data from resumes (PDF/DOCX)
2. Parse job descriptions into structured format
3. Generate ATS-friendly resumes
4. Enable iterative refinement using LLM
5. Provide ATS feedback with keyword analysis

3. System Architecture

Resume & Job Input → Text Extraction → LLM Processing → JSON → Resume Tailoring → ATS Feedback → Iterative Refinement → DOCX Output

4. Technologies Used

1. Python
2. Ollama (local LLM runtime)
3. Gemma model
4. Streamlit UI
5. pdfplumber, python-docx
6. requests

5. Implementation Details

5.1 Resume Extraction

Resume data is extracted from PDF/DOCX using parsing libraries and converted into structured JSON using LLM prompts.

5.2 Job Description Parsing

Job descriptions are processed to extract requirements, responsibilities, and keywords using LLM.

5.3 Resume Tailoring

The LLM rewrites and optimizes resume content by aligning it with job requirements and injecting relevant keywords.

5.4 Document Generation

Structured JSON output is converted into a well-formatted DOCX file using python-docx.

5.5 Iterative Refinement

Users can edit resumes and refine them using LLM, enabling continuous improvement.

6. ATS Feedback System

The system evaluates resume-job alignment and provides:

1. Match Score (0-100%)
2. Missing Keywords
3. Improvement Suggestions

7. Challenges and Solutions

1. LLM output inconsistency → fallback parsing
2. Markdown formatting → cleaning layer
3. JSON parsing failures → defensive coding
4. PDF extraction issues → preprocessing checks

8. Results

The system successfully generates personalized resumes, improves keyword alignment, and supports iterative enhancements.

9. Limitations

1. Small models may not follow strict formatting
2. ATS scoring is approximate
3. Local deployment only

10. Future Enhancements

1. Use advanced LLMs for better accuracy
2. Add resume templates
3. Deploy as web application
4. Integrate LinkedIn parsing

11. Conclusion

This project demonstrates a practical application of LLMs in resume automation. It fulfills all requirements of the capstone and provides a scalable solution for resume optimization.